

TAYLOR PENTZ

Senior Software Engineer

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SUMMARY

Senior Software Engineer with 9+ years of experience designing and delivering scalable, secure, and production-grade software systems with a strong focus on AI/ML-enabled applications. Extensive experience integrating machine learning models into real-world production environments, collaborating with cross-functional teams, and aligning technical delivery with regulatory and governance requirements. Proven background supporting data-driven platforms, implementing responsible AI practices, and ensuring transparency, explainability, and compliance in regulated domains. Adept at translating complex AI and trade-related business requirements into maintainable, high-performance software solutions. Recognized for ownership, technical leadership, and mentoring while remaining hands-on across the full delivery lifecycle.

SKILLS

Python	JavaScript	TypeScript	Machine Learning Model Integration
AI/ML Frameworks	Model Deployment	Responsible AI	Model Governance
Explainability	React	Angular	HTML5
CSS3	Accessibility (WCAG)	Responsive Design	REST APIs
Secure API Integration	AI Service Integration	AWS	Azure
CI/CD	Git	Jenkins	Production Monitoring
Audit Readiness	Regulated Environments	Governance Controls	

EXPERIENCE

Senior Software Engineer

SumatoSoft

April 2022 - December 2025

- Led the design and development of production-grade applications integrating AI/ML models using modern frameworks to support data-driven decision-making.
- Implemented end-to-end workflows for deploying, monitoring, and scaling machine learning models in secure cloud environments.
- Collaborated with data scientists and product stakeholders to translate AI research outputs into reliable, user-facing software features.
- Built governance-aware architectures that supported model versioning, traceability, and explainability requirements.
- Contributed to responsible AI initiatives by embedding transparency, bias awareness, and monitoring into application design.
- Integrated AI services with RESTful APIs to ensure secure and performant data exchange across systems.
- Supported compliance-aligned solutions relevant to international trade and government-facing programs.
- Optimized frontend and backend performance to ensure AI-powered features met strict reliability and availability standards.
- Worked closely with distributed teams to deliver scalable solutions aligned with regulatory expectations.
- Led code reviews focused on maintainability, security, and AI lifecycle best practices.
- Mentored engineers on production AI integration patterns and long-term system sustainability.

Software Engineer

Hexaview Technologies Inc.

July 2018 - March 2022

- Developed and enhanced software solutions that integrated analytical and machine learning components into enterprise platforms.
- Supported production deployment of AI-enabled features with a focus on reliability and operational monitoring.
- Collaborated with cross-functional teams to deliver systems aligned with government and international trade workflows.
- Implemented secure API integrations to connect AI services with frontend and backend applications.
- Contributed to data governance and validation processes supporting regulated environments.
- Improved system performance and scalability for data-intensive applications.
- Participated in architecture discussions emphasizing explainability and auditability of automated decision systems.
- Actively contributed to peer reviews and knowledge sharing around AI integration best practices.
- Ensured delivered features met quality, security, and compliance standards.

Software Developer

SimbirSoft Company

March 2015 - June 2018

- Built and maintained web applications supporting data processing and analytical workflows.
- Collaborated with backend teams to integrate algorithmic and rules-based decision logic into applications.
- Implemented secure data handling practices aligned with enterprise and regulatory requirements.
- Contributed to early-stage adoption of machine learning–assisted features within business applications.
- Focused on maintainable code and scalable architectures for long-term growth.
- Supported testing and validation processes to ensure accuracy and reliability of data-driven features.
- Worked in distributed teams delivering production-ready software solutions.

Software Developer Intern

Microsoft

August 2014 - February 2015

- Supported development of internal applications with exposure to large-scale, data-driven systems.
- Assisted in implementing features that integrated analytical services into web applications.
- Worked with senior engineers to follow secure coding and quality standards.
- Gained foundational experience in enterprise software engineering, compliance, and production operations.
- Contributed to testing, documentation, and continuous improvement efforts.

EDUCATION

Bachelor of Computer Science

The University of North Carolina at Chapel Hill

2010 - 2014